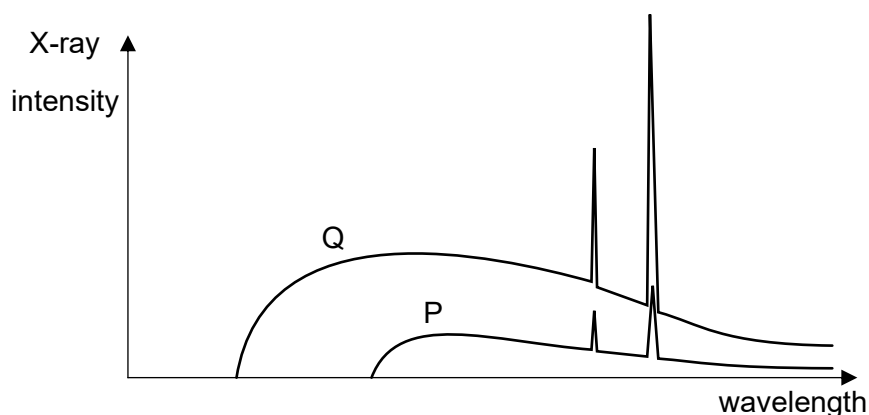


28. In the figure below, graph P shows an X-ray spectrum from an X-ray tube.



What are the possible changes that can be made to obtain graph Q?

- A Change the target metal.
- B Increase the filament current of the cathode.
- C Reduce the distance between the cathode and the target.
- D Increase the potential difference between the cathode and target.

Ans: D

Increasing the p.d. mean that the bombarding electrons will have higher ke . This will result in a smaller minimum wavelength of X-ray produced when the electrons strike the metal target. The overall intensity will also increase as the chance of emitting X-ray photons will be higher due to the higher energy electrons striking the target.

29 Which of the following statements is true?